

# Module 2: Infrastructure Security

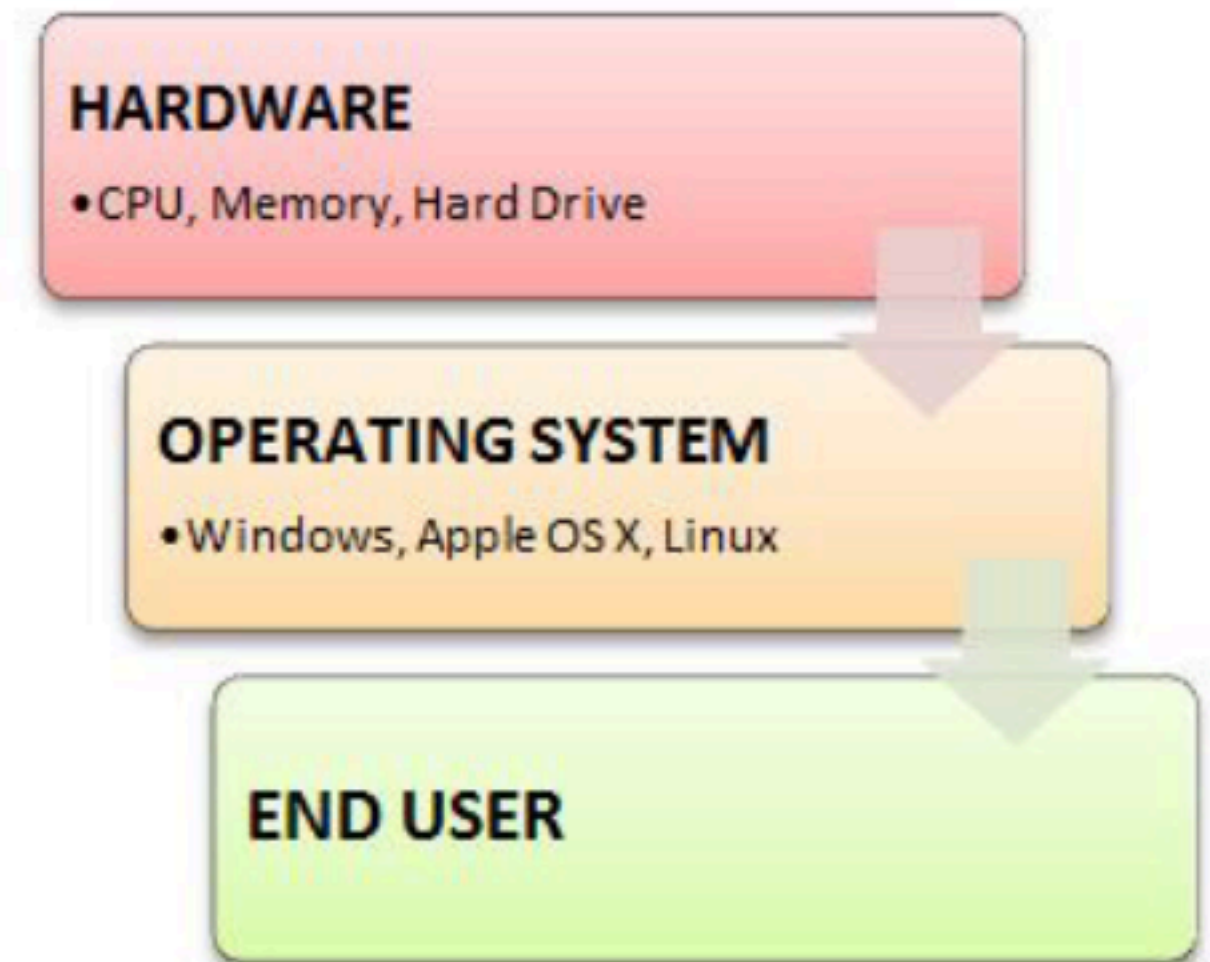


# Topic 1: Operation System Security



# Introduction To Operating Systems

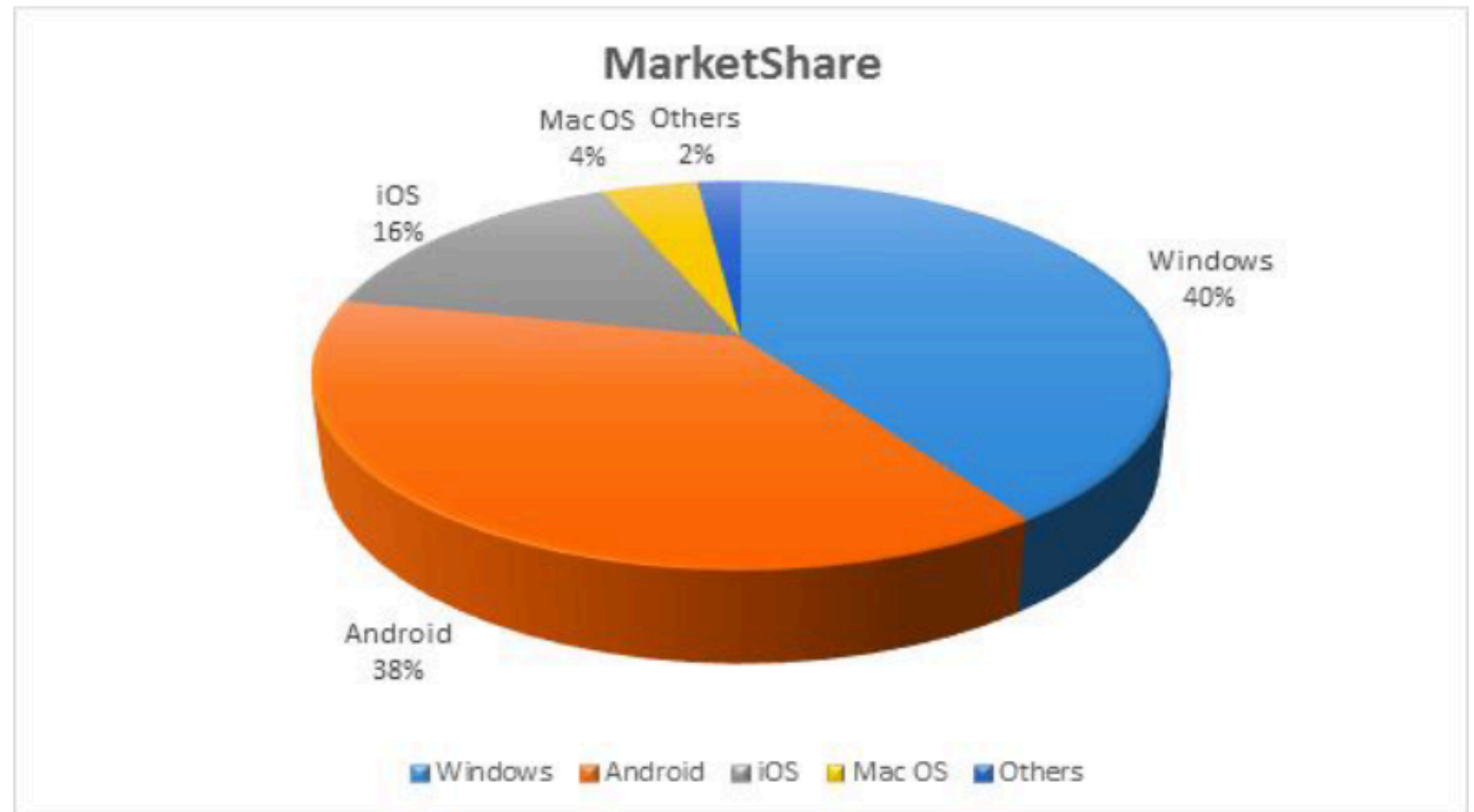
- Operating Systems (OS) are intermediate level between computer hardware and end user.
- OS acts as orchestrator that controls execution of user's program and manages the usage of hardware and how programs should use the hardware.
- OS helps us to use computer without knowing the hardware programming language.



# Introduction To Operating Systems

There are many operating systems in the market as:

- Windows
- Mac OS
- Linux
- Android
- IOS
- Chrome OS
- Others



# Introduction To Operating Systems

## Operating systems must ensure:

- **Confidentiality:** System is only accessed by authorized users & data of each user is separate from other user.
- **Integrity:** Prevent unauthorized users or applications from modifying available data on both disk & memory.
- **Availability:** System is available and ready to be used whenever needed by authorized users.
- **Authenticity:** system can verify the identity of the user before accessing it.



# Common OS Threats

- Malicious applications (viruses, worms, trojans, ransomwares, etc.)
- Data loss
- Denial of service
- System crash
- Unauthorized data access
- Unauthorized resources consumption
- Zero days
- Privilege escalation

# How to Secure OS?

## Step 1

**Choose a secure OS known and trusted.**

## Step 2

**Ensure secure design and installation process through:**

- Using secure hardware
- Use secure methods for installing OS
- Enable security features

## Step 3

**Apply the controls that should help in securing the OS**

## Step 4

**Use Evaluation criteria or checklist that ensures all the required steps are performed.**



# How to Secure OS?

- Use effective, well known and trusted antivirus
- Use Firewalls
- Enforce user authentication
- Ensure performing vulnerability and patch management for the system
- Apply encryption
- Enforce file permissions
- Apply hardening for the OS
- Activate logging and monitoring for the desired security events (ex: failed logins, logoffs, change in permissions, etc.)
- Security audits
- Perform vulnerability assessments
- Do penetration testing



# System Hardening

System hardening is the process of securing operating system by reducing its vulnerabilities or weak areas in the system.

The system hardening involves applying many configurations and removing unnecessary services or applications from the system to minimize areas that can be attacked (attack surface).

Hardening can be applied on Operating systems, Databases, applications, network devices.

# System Hardening

## **System hardening activities:**

- Use standard baseline (minimum security configurations to be applied on system)
- Remove unnecessary applications
- Block unnecessary ports and services
- Disabled unneeded and unused accounts
- Change default passwords on systems
- Rename default accounts
- Apply security policies

# Patch Management

The patch is a fix that is used to close a security gap or solve a detected problem in the system.

Usually, all vendors release patches to close security vulnerabilities that were detected by their teams or attacked by malicious actors.

Patch management means ensuring systems and applications are up to date with latest approved security patches and updates.

Patch management is the process of regularly installing security patches and updating system and applications in timely manner.

# Patch Management

To have effective patch management, it should have a process that:

- 1- Patch Identification:** Regularly checking vendor notifications and security bulletins for released patches.
- 2- Patch Acquisition:** Obtaining needed patches according to security needs.
- 3- Testing and validation:** Validate patch effectiveness on testing. environment to ensure it has no impact on environment.
- 4- Deployment:** Deploy patch of target system.
- 5- Monitoring:** Ensure the system is properly operating and all services are working normally.